

HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

DoseTwin

Test the dose on your twin before you inject it.

A personal digital twin for insulin-treated diabetes, validated on type 1. It checks tonight for lows at bedtime, warns before lows by day, and tests every meal dose, using CGM, pen, wearable and health-record data.

Every insulin dose is a manual guess

9.4 lakh

people in India live with type 1 diabetes; about 3 lakh are under 20.

4–5×

insulin pen injections a day. Most families cannot afford pumps or automated delivery.

After

Affordable CGM sensors alarm only after glucose is already low, often at night.

Each dose depends on a carb guess, clinic ratios set months ago, and the day's activity. Too much insulin causes hypoglycaemia; too little causes long-term damage.

Why the bolus calculator gets it wrong

What the calculator assumes	What actually happens
Carbs are counted correctly	Counts are off by 20–50%, with a personal bias
Clinic ratios are right	Carb ratio and correction factor drift; reviews are months apart
Every day is the same	Exercise, short nights, the dawn rise and kidney function change the need
Insulin on board is linear	Absorption differs by person, dose and timing

A twin that knows **this** person's physiology can answer the question the calculator cannot: **what will this dose do over the next five hours?**

DoseTwin does three jobs

ADVERSE EVENT

Checks the night at bedtime

At 23:00 the twin simulates the night. If a low is likely, it sizes a bedtime snack before the person falls asleep.

ADVERSE EVENT

Warns before daytime lows

Twin forecast plus CGM trend, about 26 min ahead, while there is time to eat 15 g.

PERSONAL TREATMENT

Tests every meal dose

Simulates 5 hours for each half-unit dose and picks the lowest-risk dose above a safety floor.

Inputs: CGM · smart-pen doses · meal log · steps, heart rate and sleep · synthetic EHR (diagnoses, labs incl. HbA1c and eGFR, genetics, prescriptions).

Architecture

DEVICE DATA

CGM, every 5 min

Smart-pen doses

Meal log

Steps, heart rate, sleep

Synthetic EHR: HbA1c,
eGFR, hypo history, Rx



PERSONAL TWIN

EHR prior: prescriptions, HbA1c and eGFR set the starting physiology

MAP fit on the patient's own 5-hour forecasts, about 1.5 s

Observer locks the twin to every CGM reading: insulin and carbs on board, activity, last night's sleep



Bedtime check

8-h night forecast at 23:00;
snack size

Low alert

Twin 40-min forecast + CGM trend

Dose advisor

5-h what-if per half unit; floor 70
(80 with hypo history)

Python · NumPy · Numba · SciPy · LightGBM baseline · browser twin in JavaScript (parity-tested to 1e-6 mg/dL)

AI and model details

Component	What it is
Twin model	10-state extended Bergman model: SC insulin, carb absorption, activity from steps + heart rate, sleep-debt insulin resistance, dawn effect
Personalisation	EHR prior (weight, prescriptions, HbA1c, eGFR), then MAP fit on 5–300 min forecast error
Synchronisation	Fixed-gain observer on CGM; unexplained-rate state captures surprises
Dose advisor	Kovatchev risk minimisation; rejects doses whose pessimistic path dips below 70 (80 with severe-hypo history)
Bedtime check	8-h nominal and pessimistic night forecast at 23:00; flags below 84 mg/dL; snack sized on the twin
Low alert	Hybrid score: twin forecast minimum + 30-min CGM trend; alerts below 65
Baseline	LightGBM forecaster, low classifier and night classifier, same inputs, trained on 30 other patients

The twin is interpretable: every parameter has a physiological meaning a clinician can review, and it can answer **what if** questions that a pure ML forecaster cannot.

DT

DoseTwin

Digital twin for insulin-treated diabetes

Patients

Dose check

Evidence

Method

How to use

Synthetic data

In-silico study · not a medical device

30 PATIENTS · HIGHEST RISK FIRST

Adult, 35 y, M

adult#007 · HbA1c 5.5%

<7010.4%

TIR87%

Low likely tonight

Adult, 26 y, F

adult#006 · HbA1c 5.5%

<7010.1%

TIR83%

Low likely tonight

Child, 10 y, M

child#005 · HbA1c 5.6%

<709.9%

TIR85%

Low likely tonight

Child, 8 y, F

child#003 · HbA1c 6.1%

<708.6%

TIR78%

Low likely tonight

Child, 12 y, F

child#010 · HbA1c 6.7%

<706.4%

TIR68%

Low likely tonight

Child, 8 y, M

child#006 · HbA1c 5.7%

<708.5%

TIR73%

Low likely tonight

Child, 10 y, F

child#008 · HbA1c 7.8%

<704.7%

TIR56%

Low likely tonight

Child, 12 y, F

child#004 · HbA1c 5.7%

<706.4%

TIR91%

Low likely tonight

Child, 7 y, F

child#009 · HbA1c 5.7%

<705.1%

TIR88%

Low likely tonight

Child, 9 y, M

child#002 · HbA1c 6%

<706.8%

TIR85%

Low likely tonight

Child, 9 y, F

child#001 · HbA1c 6%

<704.1%

TIR81%

Low likely tonight

Adult, 66 y, M

adult#004 · HbA1c 5.9%

<704.9%

TIR81%

Low likely tonight

Adult, 65 y, M

adult#002 · HbA1c 5.6%

<704.2%

TIR94%

Low likely tonight

Adult, 27 y, M

adult#003 · HbA1c 7.1%

<702.1%

TIR75%

Low likely tonight

87%

Time in range 70–180

On target · >70%

10.4%

Below 70

Above target · <4%

0.5%

Below 54

On target · <1%

24.1%

Below 70, 00–07 h

Above target · <4%

30%

Variability (CV)

On target · ≤36%

22

Low episodes, 7 days

Above target · fewer is better

GLUCOSE PROFILE, LAST 7 DAYS (MEDIAN, 25–75% AND 10–90% RANGES)

TONIGHT · BEDTIME CHECK AT 23:00 (END OF DAY 7)

Low likely tonight.

The pessimistic forecast reaches 67 mg/dL around 01:45, 2.8 h after bedtime (alert line 84 mg/dL). The twin suggests a **20 g** bedtime snack, the smallest that keeps its pessimistic night above 80 mg/dL.

VIRTUAL PATIENT REPLAYS DAY 7 UNDER THE TRIAL SETTINGS

Twin prediction for day 7: time in range **96% → 96%** (+0), time below 70 **4.2% → 4.2%** (+0.0), lowest 67 → 67 mg/dL.

Same meals, carb counts and steps as the recorded day; doses recomputed with the trial settings. This is the twin's prediction, not a measurement.

DIAGNOSES

- Type 1 diabetes mellitus E10
- Background diabetic retinopathy E10.3

LABS

HbA1c5.5%

Fasting C-peptide0.14 nmol/L

eGFR82 mL/min

TSH2.9 mIU/L

LDL147 mg/dL

Urine ACR12 mg/g

GENETICS

HLA-DRDR4/X

GAD65 at diagnosispositive

MEDICATIONS

- Insulin glargine: 38 U once daily
- Insulin aspart (pen): 1 U per 22 g carbohydrate; 1 U lowers glucose 23 mg/dL

WEARABLES, LAST 7 DAYS

Sleep, average6.7 h

Nights under 6 h1

Sleep debt going into tonight0.0 h

Resting heart rate62 bpm

Days with exercise seen only by heart rate1

Synthetic record. The twin uses weight, insulin prescriptions, HbA1c, eGFR (slower insulin clearance) and the severe-hypo history (higher safety floor, earlier alerts); the rest is context for the clinician.

SETTINGS REVIEW

Setting	Current	Twin estimate	Suggested
Carb ratio (g per U)	21.5	42.5	25.8 +20%
Correction (mg/dL per U)	23	37	27 +20%
Basal (U per day)	37.6	36.3	36.3 -3%

Suggestions move at most 20% per review, in the direction the twin indicates. When the twin and the record differ more than twofold, the twin is flagged as low confidence and the setting is kept.

Carb ratio

1 U : 25.8 g

Correction

27 mg/dL

The doctor's view

Patients ranked by hypoglycaemia risk, including tonight's bedtime check, with the record, wearables, glucose profile and tonight's forecast.

Move a setting and the virtual patient replays a real day under it.

Suggestions change at most 20% per review and are flagged when the twin is unsure.



Live demo

Drag the dose. The twin re-simulates the next 5 hours in the browser.

Dinner for a 12-year-old: the calculator says 3.5 U; the twin says 1.5 U.

Real outcome: 60 min below 70 with 3.5 U, none with 1.5 U (peak 242 mg/dL).

Example meal picked by a fixed, disclosed rule. Cohort results follow.

How we tested it without fooling ourselves

Hidden truth

30 virtual patients from the published UVA/Padova T1D model. The twin uses a different model and never sees it.

Realistic life

Late Indian dinners, 30% carb-count error, missed boluses, mis-set clinic ratios, exercise, short nights, kidney disease, impaired awareness, sensor noise.

Held-out cohorts

All settings frozen on a development cohort. Results come from 3 fresh cohorts: 90 patient-weeks, 529 nights.

Paired arms

Every arm relives the same 7 days. Ablations refit the twin. Bootstrap CIs and Wilcoxon tests.

Fewer lows, especially at night

-77%

time below 70 mg/dL (3.43% → 0.79%)

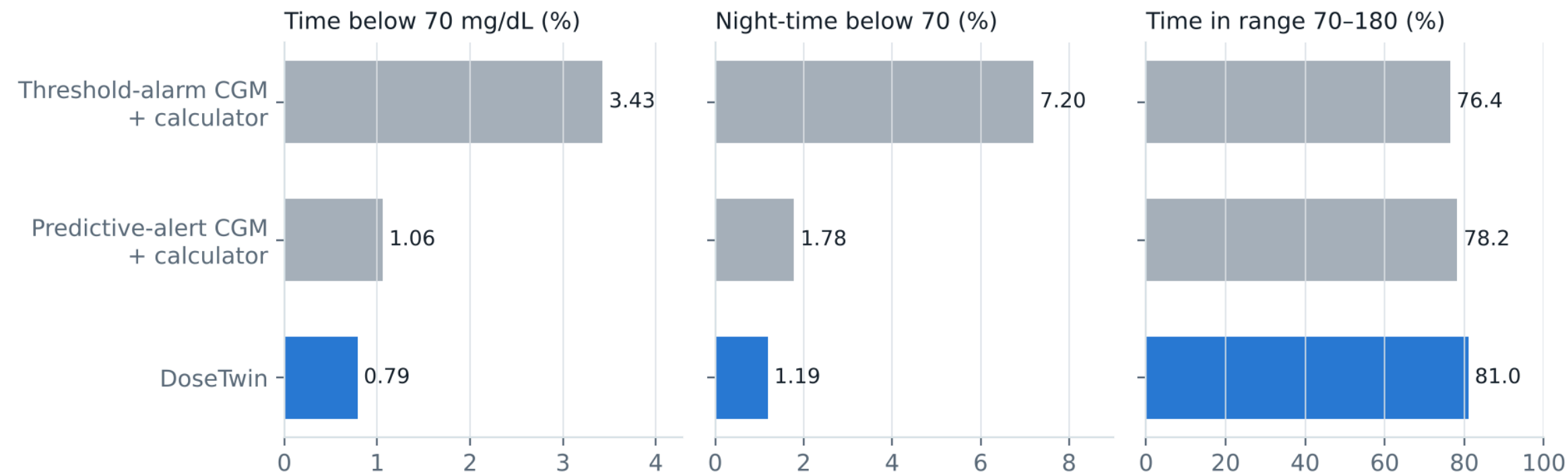
-83%

night-time below 70 (7.2% → 1.2%)

+4.6 pts

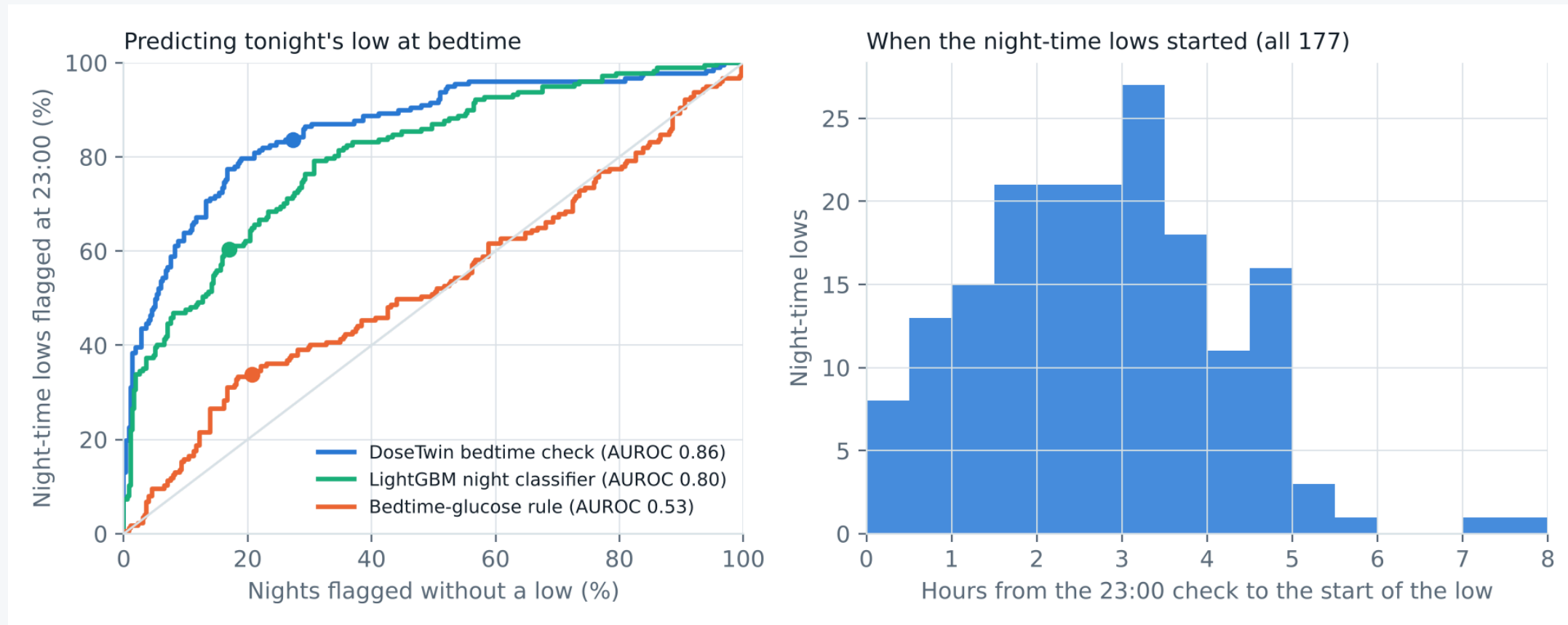
time in range (95% CI 2.1 to 7.5)

Outcomes on identical days, held-out cohorts (30 patients × 3)



Full DoseTwin vs threshold-alarm CGM + bolus calculator, all $p < 0.001$. Vs premium predictive-alert CGM: TIR +2.8 ($p = 0.05$), time below 70 1.06% → 0.79% ($p = 0.11$), night 1.78% → 1.19% ($p = 0.08$): not significant.

Tonight's low, predicted at bedtime



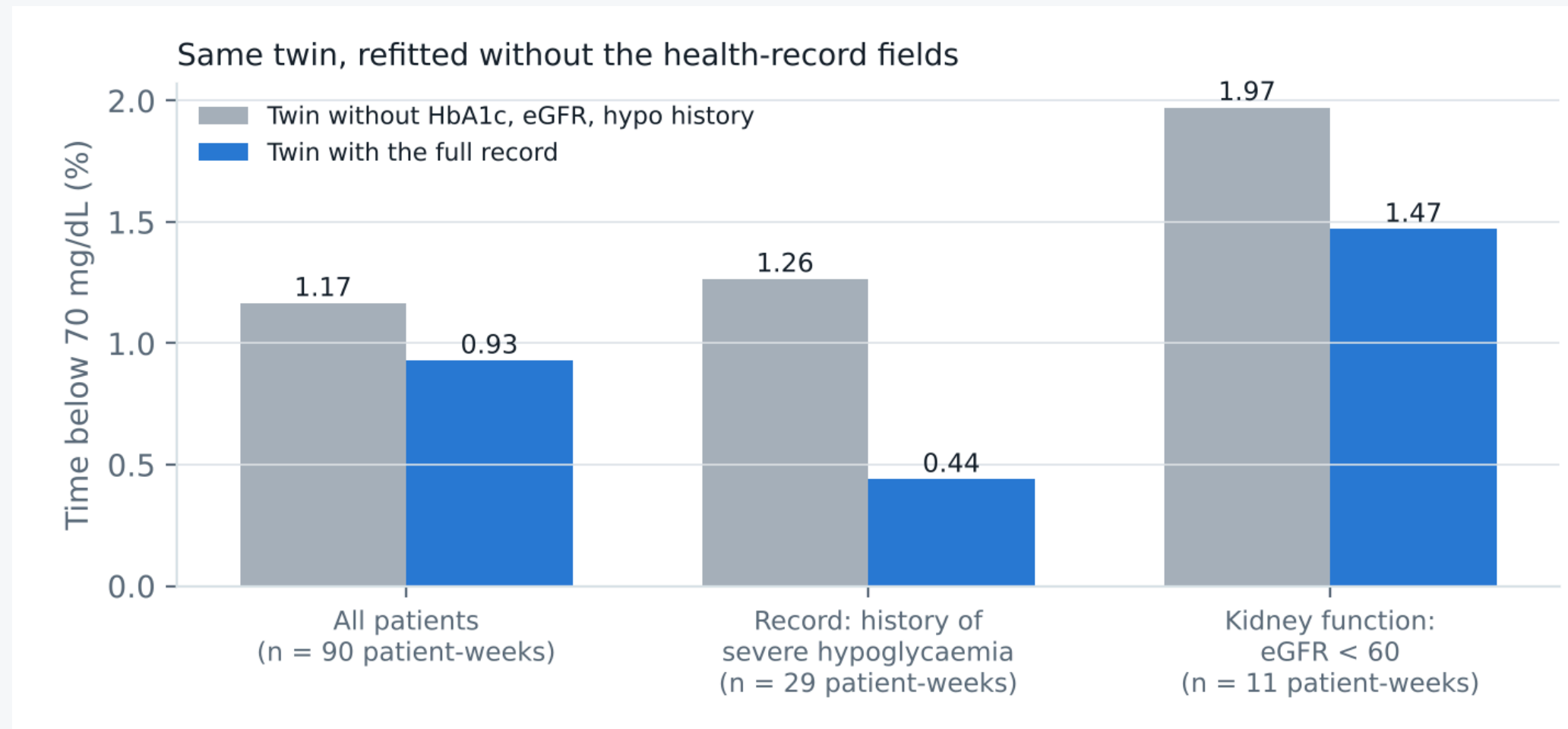
AUROC **0.86** vs 0.80 for LightGBM (+0.06, 95% CI 0.01–0.11) and 0.53 for the bedtime-glucose rule.

Flags **84%** of night lows, a median **2.5 h** before they start, before the person falls asleep.

A twin-sized snack cut night-time below 70 by **27%** (1.63% → 1.19%, $p = 0.03$), for 2.5 snacks a week.

529 held-out nights, 177 with a low ($< 70 \text{ mg/dL} \geq 15 \text{ min}$, 23:00–07:00). Thresholds frozen on the development cohort. It also flags 27% of nights that stay safe.

The health record changes the twin's decisions



eGFR: weak kidneys clear insulin slowly, so the twin expects each unit to act more strongly.

Severe-hypo history: dose floor 80 instead of 70 mg/dL, alerts 7 mg/dL earlier.

HbA1c: sets the twin's fasting glucose.

Refitted without them: time below 70 **0.93%** → **1.17%** ($p = 0.01$); with severe-hypo history **0.44%** → **1.26%**.

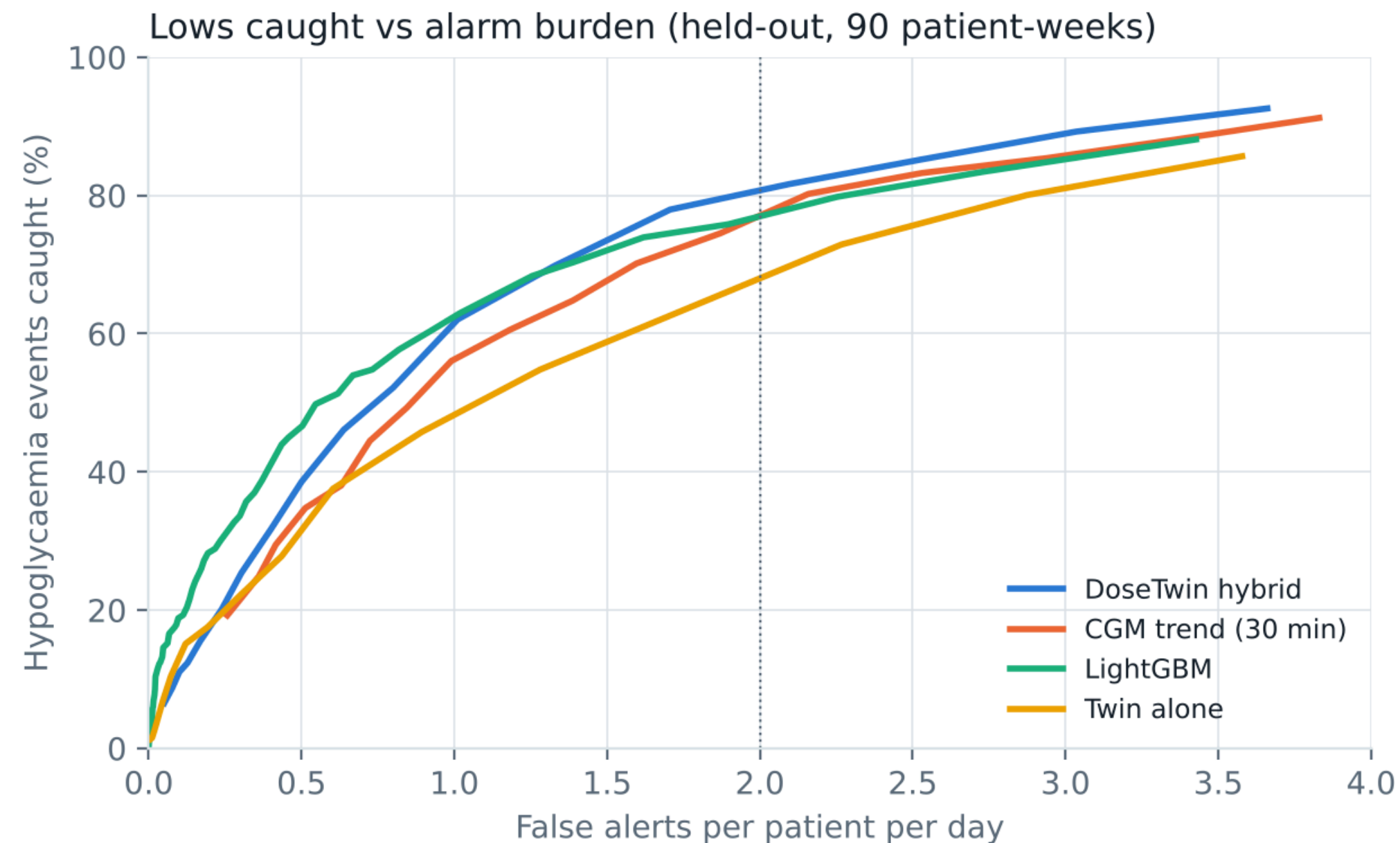
Cost: 1.3 points of time in range. Subgroups are small (29 and 11 patient-weeks). Thyroid, coeliac and genetic fields are shown to the clinician but not used.

Catching lows early

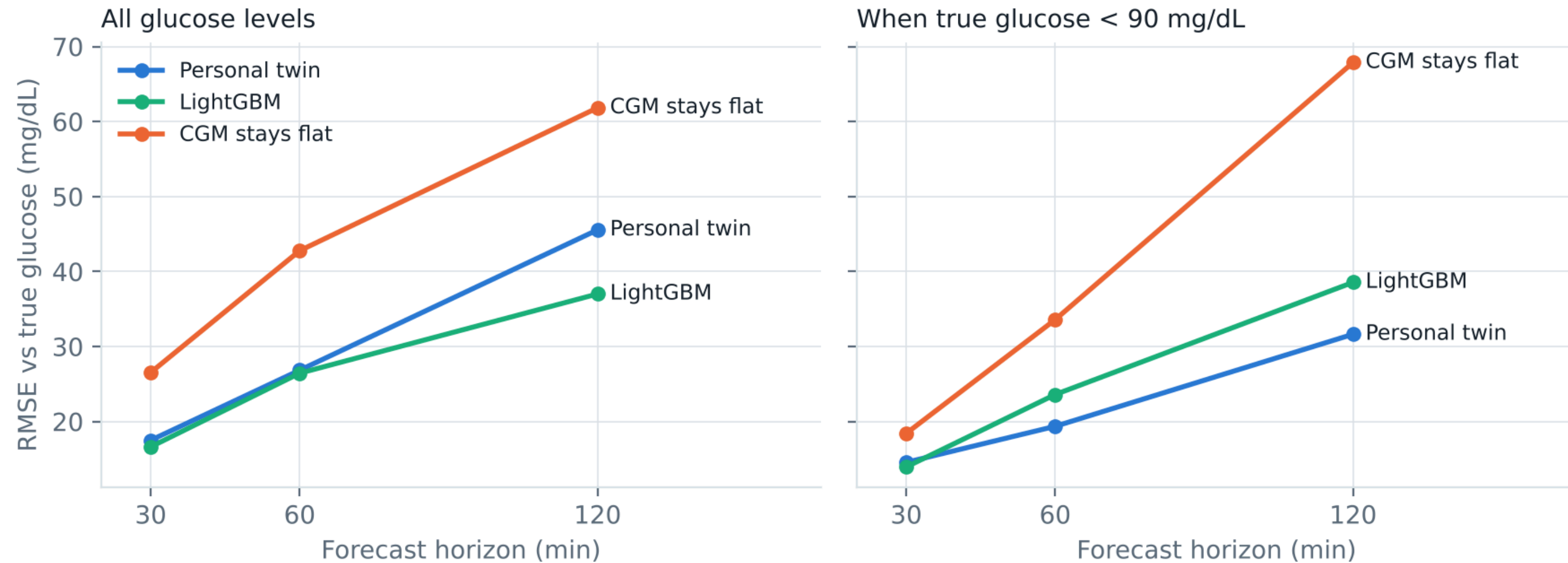
At ≤ 2 false alerts a day, DoseTwin caught **78%** of lows with a median 26 min warning.

CGM trend line: 75%. LightGBM trained on 30 patients: 76%.

LightGBM wins at ≤ 1 alert a day and warns earlier. The twin alone lost to the trend line; the hybrid edges both.



Most accurate where it matters: near lows



Below 90 mg/dL, 60-min error is 19.4 mg/dL for the twin vs 23.6 for LightGBM. Overall, LightGBM is as good or better, and better at 2 h. The twin learns from 7 days of one person's data and can simulate doses; LightGBM cannot.

What this does not prove yet

Limitation	What we found
Virtual, type 1 only	Real-patient validation, then type 2 on insulin, are next; cohort enriched for kidney disease and impaired awareness
No 4–7 h warning	Bedtime check warns a median 2.5 h ahead; it also flags 27% of nights that stay safe
Heart rate and sleep	No measurable benefit: bedtime AUROC 0.87 without them vs 0.86 with
Vs premium CGM	Gains in every measure, none significant at 0.05
Settings recovery	Twin carb ratio closer to truth in 54% of cases, correction factor in 44%; changes capped at 20% and clinician-approved
Easy cohort	Standard-care TIR ~76%, better than real life

Premium protection on an affordable sensor

For families

Earlier warnings at night and a second opinion on every dose, using the phone and sensor they already have.

For clinics

A twin-based review of carb ratio, correction factor and basal between visits, with every suggestion explained.

Roadmap

Real-data validation · CGM and smart-pen connectors · ABDM/FHIR export · paediatric clinic pilot

Built to be checked

Item	Detail
Stack	Python 3.11, NumPy, Numba, SciPy, pandas, LightGBM, scikit-learn, Matplotlib; vanilla JavaScript demo
Tests	17 tests: simulator parity (< 0.5 mg/dL), twin, advisor safety, EHR effects, browser parity ($< 1e-6$)
Reproduce	One command, about 25 min on a laptop CPU; Dockerfile included
Speed	Twin fit about 1.5 s per patient; dose search is real-time in the browser
Licence	MIT; simulator parameters from simglucose (MIT)
Data	No real patient data used; nothing personal stored

Test the dose on your twin before you inject it.

83% less night-time hypoglycaemia than an affordable threshold-alarm CGM, and tonight's low predicted at bedtime with AUROC 0.86, in a held-out in-silico study.